Robust and Accurate System for Hand Pose Data Acquisition of Dexterous Manipulation

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***Abstract*—Although learning techniques have largely achieved success in robotic manipulation, dexterous manipulation remains challenging in that even the data collection is not trivial. In this work, we propose a system to acquire human demonstration data for efficient learning of dexterous manipulation using a hand tracking platform, the visual-inertial skeleton tracking (VIST) [1] system. The human demonstrator wears the VIST glove and performs manipulation tasks in the real world, where the system tracks the hand motion accurately and robustly. Unlike other data acquisition frameworks based on vision (e.g., stereo vision) or inertial measurements (e.g., inertial measurement units (IMUs)), our system fuses both of the sensing modalities, which renders it robust to visual occlusion and electromagnetic interference. Moreover, the system enables more versatile tasks (e.g., in-hand object manipulation) compared to soft sensor-based tracking, which cannot distinguish contact and deformation, and exoskeleton-based tracking, which is too bulky to perform fully dexterous tasks. In addition to the hand motion, either contact sensors (e.g., force sensitive resistor (FSR)) or contact forces estimated from physically accurate simulators [2, 3] can be integrated to enrich the data with the tactile information from which robotic dexterous manipulation would benefit. We present samples of data collected in dexterous manipulation tasks that are challenging with existing methods (e.g., visually occluded and contact-rich), which demonstrates the effectiveness and versatility of our data acquisition framework.**

I. Introduction

Dexterous manipulation is an appealing ability for robots to be generally used for various tasks but still remains significantly challenging due to the high-dimensional state and contact-rich nature of the manipulation. Although deep learning techniques utilizing expert data (e.g., imitation learning (IL) of human demonstration) have evolved and proved to be surprisingly promising in many robotic tasks, for dexterous manipulation, it is not straightforward to even obtain the human data for its training.

One main approach for the data acquisition is using vision (e.g., RGB or RGB-D) [4, 5, 6, 7]. Teleoperation systems of a human hand and a robotic hand were proposed in [4] and [5] using an RGB-D camera and four monochrome cameras, respectively. In [6], adopting an RGB-D camera to capture the human hand motion, robotic hands were teleoperated in a simulator rather than in the real world. Similarly, human hand demonstrations were recorded using an RGB-D camera in [7] and transferred to a robotic hand. These vision-based frameworks are fundamentally susceptible to visual occlusion since they cannot estimate the unseen parts of the hand. Another approach employed inertial measurement units (IMUs) in a glove to obtain the human hand motion data, along with tactile sensors to record the touch information [8]. However, IMUs fail in the presence of electromagnetic interference (e.g., near power tools) that can often arise from manipulation tasks (e.g., using an electric drill). Alternatively, soft sensors were exploited in [9] to track hand configuration, but the soft sensors cannot differentiate between measurements by elongation/compression and measurements by contact, and thus are unable to handle object manipulation with contact. One last approach uses an exoskeleton glove to acquire human demonstration data [10, 11]. Although they are advantageous in that the exoskeleton glove can provide force feedback to the user for a more immersive demonstration, the design is bulky and it generates friction, which limits dexterous manipulation tasks to simple tasks such as grasping.



Fig. 1. VIST [1] exploited to acquire human demonstration data for dexterous manipulation.

In this work, we propose a system to acquire demonstration data of human experts for efficient learning of dexterous manipulation, which is built upon a hand tracking platform, the visual-inertial skeleton tracking (VIST) [1] system shown in Fig. 1. We obtain the data while the user wears the VIST glove and performs manipulation tasks in the physical world. By fusing inertial and visual information in a tightly-coupled (TC) manner for tracking, our system is robust to visual occlusion (e.g., self-occlusion and occlusion by an in-hand object) and electromagnetic interference from manipulated objects (e.g., electric drill). The system is also robust to contact that nat-

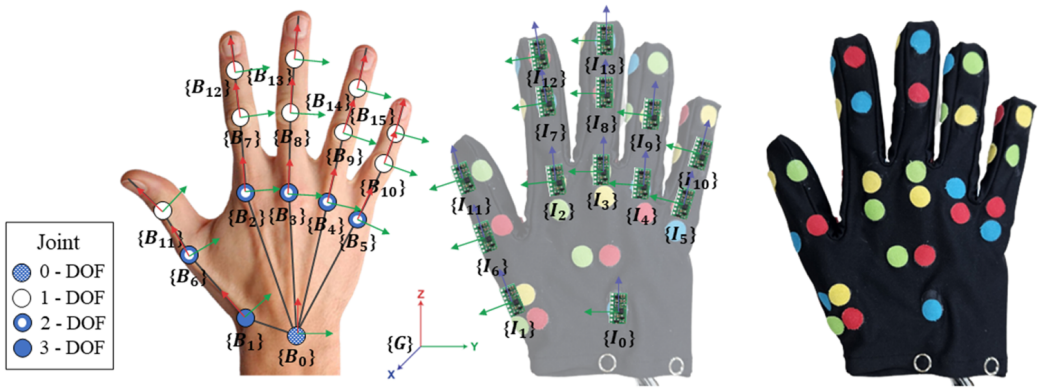


Fig. 2. Hand and glove models with coordinate frames.

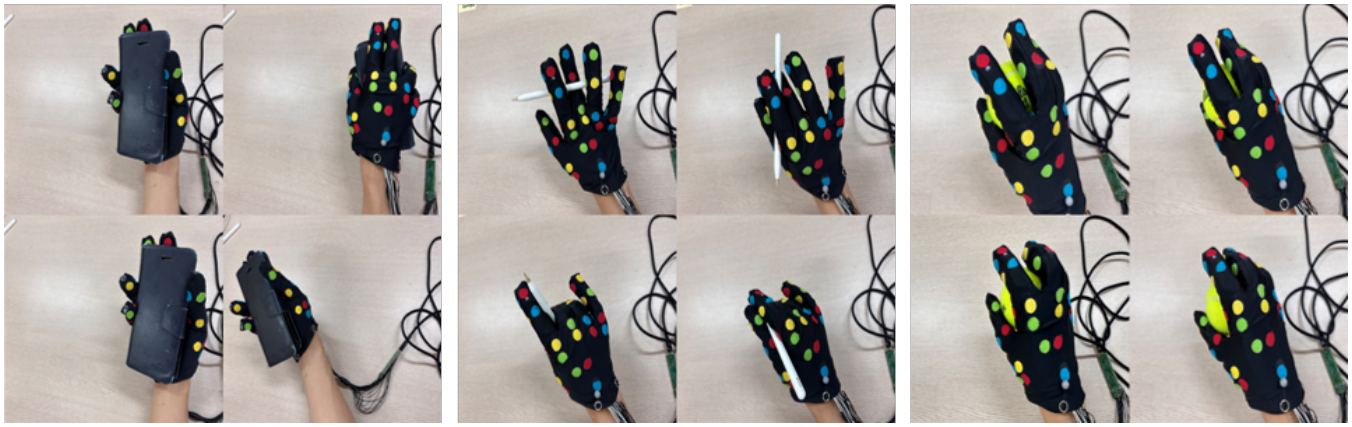


Fig. 3. Scenarios for dexterous manipulation: manipulating a large object (left); spinning a thin pen (middle); and rolling a ball with palm facing downward (right). The order of each pose is from left to right, top to bottom.

urally occurs during in-hand object manipulation, unlike soft sensor-based approaches. And the compact and lightweight design of the system enables the human expert to perform a fully dexterous manipulation (e.g., pen spinning) that is impossible in bulky exoskeleton-type data acquisition methods. The system can easily be extended to augment the data with contact information either by integrating contact sensor modalities (e.g., force sensitive resistor (FSR)) or by estimating contact forces from physically accurate simulators [3]. Since robotic dexterous manipulation benefits a lot from the sense of touch, the inclusion of touch data would significantly enhance the versatility of the data for in-hand object manipulation. We present several samples of data collected in dexterous manipulation that are difficult to acquire with existing methods (e.g., due to visual occlusion and rich contact), which demonstrates the effectiveness and versatility of our data acquisition framework.

II. System Description

To obtain the hand motion data of the user, we use the VIST system [1] as a hand tracking framework. The VIST system adopts a segment-joint skeleton model and estimates the poses of 16 joints $\{B_i\}$ $(i = 0, 1, 2, ..., 15)$ of human hands using a sensor glove. The glove has two layers: 1) the inner layer has 14 IMUs (Pololu, LSM6DS33) attached to 14 of 16 segments $\{I_j\}$ $(j = 0, 1, 2, ..., 13)$; and 2) the outer layer has 65 colored fabric markers with 4 different colors on it, as shown in figure 2. IMU data is transmitted to a PC via an MCU board (PJRC, Teensy 4.0, ARM Cortex-M7), and a stereo camera (Stereolabs, ZED mini) is used to detect the markers. The performance evaluation of the VIST system shows an RMSE mean error of 13.4mm for the five important locations (thumb fingertip, index fingertip, middle fingertip, left back of the hand, and right back of the hand) with a speed of 189 Hz. Due to the complementary use of these visual and inertial sensor modalities, our hand tracking becomes not only highly accurate but also robust enough to be used in real-world applications.

The VIST framework fuses the visual and inertial sensors in a tightly-coupled (TC) manner, enabling bidirectional information fusion: 1) from visual to inertial (e.g., IMU drift correction); and 2) from inertial to visual (i.e., IMU-aided marker correspondence search). IMU-aided marker correspondence search aims to find the correspondence of the set of stereo-matched markers to the set of IMU-predicted positions of the visual markers. The tracking deploys the extended Kalman

filter (EKF) to run the sensor fusion at a rate faster than the sensor sampling rate. The state for the filtering is defined for each segment of the hand (total 14 segments - see Fig. 2) as

$$x := [x_s; x_p] \in \mathbb{R}^{23} \tag{1}$$

$$x_s := \left[p^G_{G,I}; v^G_{G,I}; q^G_{G,I}; b_g; b_a\right] \in \mathbb{R}^{16} \tag{2}$$

$$x_p := \left[\lambda_B; q^I_{I,B}\right] \in \mathbb{R}^7 \tag{3}$$

where $x_s \in \mathbb{R}^{16}$ is the motion-related state of the segment (concatenation of the position $p^G_{G,I} \in \mathbb{R}^3$, velocity $v^G_{G,I} \in \mathbb{R}^3$, and orientation quaternion $q^G_{G,I}$ of the IMU coordinate frame $\{I\}$ in the global coordinate frame $\{G\}$, and IMU biases $b_g \in \mathbb{R}, b_a \in \mathbb{R}$), and $x_p \in \mathbb{R}^7$ is the state for the autocalibration of hand/sensor-related kinematic parameters (concatenation of hand scale parameter $\lambda_B \in \mathbb{R}^3$ and IMU misalignment quaternion $q^B_{I,B}$). By also estimating λ_B and $q^I_{I,B}$, our system is able to autocalibrate hand anatomical parameters for each user in real-time. In the prediction step, the state of the segment is propagated using the accelerometer and gyroscope measurements. Then the visual observation is processed into marker information through marker detection and IMU-aided marker correspondence search. The correction is done using this marker information and anatomical constraints (e.g., adjacent segments are connected at the pivot joints and rotational constraints of joints).

We build the system for data acquisition on the VIST framework, where the position and orientation of each hand segment are tracked real-time. In order to acquire the fully-configured data of the hand motion, we extend VIST hardware from three digits (Fig. 1) to five digits (Fig. 2). The hand tracking system can be easily extended by adding other sensors (e.g., tactile sensors such as FSR sensors) or actuators (e.g., cutaneous haptic devices [12]). For example, if FSR sensors are attached to the fingertips of the VIST sensor glove, we can obtain contact force data during in-hand manipulation. This data can be utilized to learn to perform various dexterous manipulations involving interaction with environments (e.g., object manipulation).

III. Result

We present data acquisition samples for three in-hand manipulation scenarios: 1) manipulating a large object; 2) spinning a thin pen; and 3) rolling a ball with palm facing downward, which are depicted in Fig. 3. Each scenario represents a challenging situation when acquiring data with vision-based frameworks or those using soft sensors. The first scenario is the case when there is a severe occlusion due to the manipulated object. As shown in Fig 3, the 3rd, 4th, and the 5th fingers are mainly hidden during the scenario, which can cause difficulty obtaining data for those fingers when using vision-based frameworks. The second scenario involves fast-changing and multiple contacts between the object and the hand, even on the dorsal side of the hand. This can distort the data when soft sensors are used to measure the configuration of the finger because such sensors cannot distinguish measurements by elongation/compression from measurements by contact. The last scenario is the case when the object can disturb the visual part of our system (e.g., marker detection) due to its similar color to the marker.

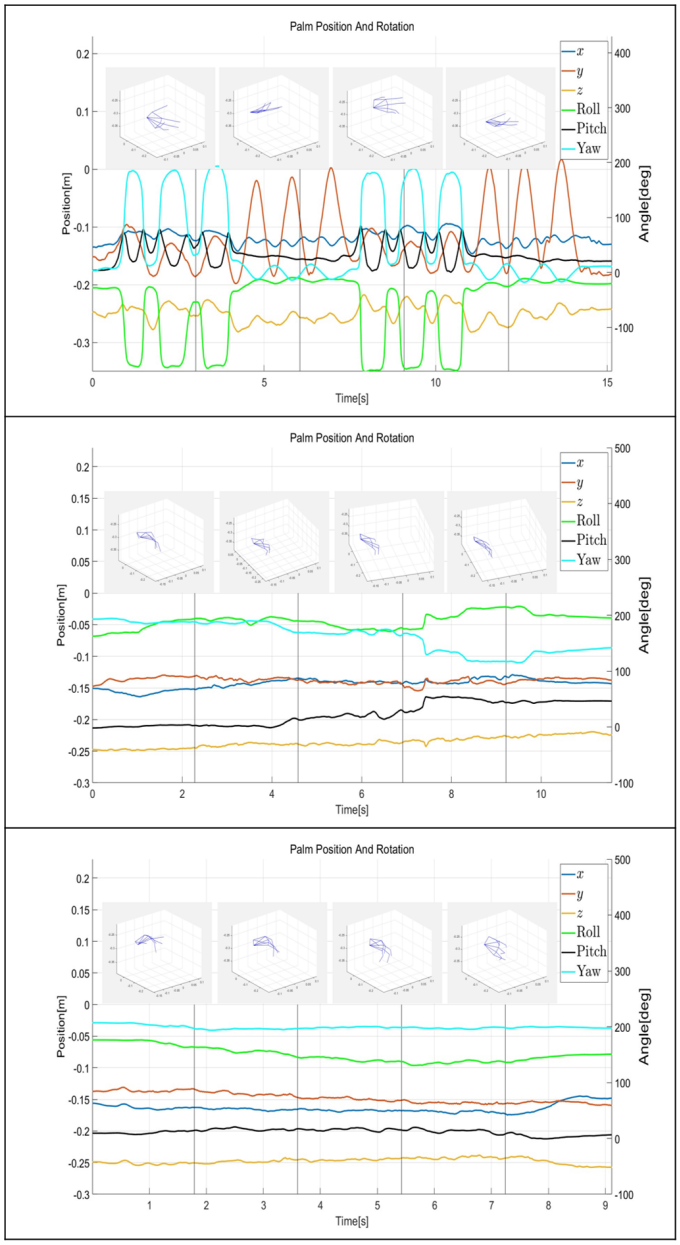


Fig. 4. Results for each scenarios: manuplating a large object (top); spinning a pen (middle); and rolling a ball with palm facing downward (bottom). The palm position and orientation plot is presented according to time and hand skeletons are shown for some timesteps.

We choose specific tasks for each scenario before acquiring the data. The task for each scenario is as follows; task for the first scenario is to flip and shake the object for three times and repeat them; task for the second scenario is to spin the pen by starting from the first pose to the last pose as shown in Fig 3; and the task for the last scenario is to roll the tennis ball of which logo is facing the palm to make it face the table.

For each scenario, the positions and rotations of 16 bodies

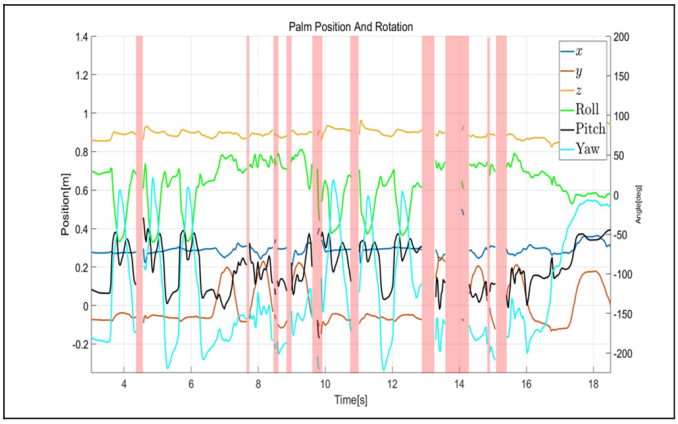


Fig. 5. Result for the first scenario obtained using 'HandTracking2.0' (Oculus). The red region represents where tracking confidence is low.

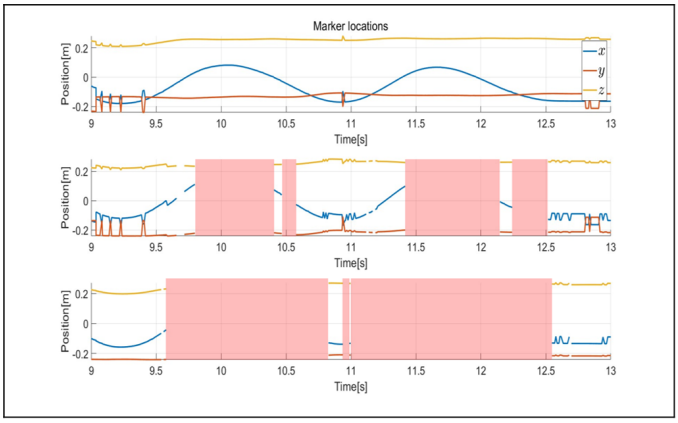


Fig. 6. Result for the first scenario obtained using 'VICON'. The red region represents where tracking is lost.

including the $1^{st}\sim5^{th}$ distales are measured for 20 to 30 [s]. The 4th and 5th distales are extrapolated from the kinematic synergies. Data acquisition is done successfully without drifting or diverging for every scenario. The data (hand skeletons of some timesteps, palm position, and palm orientation) is shown for the third scenario in Fig. 4.

We also obtain data for the first scenario using 'HandTracking2.0' (Oculus) and 'VICON'(Motion capturing system) to compare the performance for the method using only vision-based framework to our method. As shown in Fig. 5 and Fig. 6, both methods often fail to track the hand pose due to severe occlusion while our method shows robust performance as shown in Fig. 4.

IV. Conclusion

We propose a system for data acquisition of human demonstrators for efficient learning of dexterous manipulation, based on a hand tracking framework, VIST [1]. In our system, data is collected while the demonstrator wears the VIST glove and performs dexterous manipulation tasks in the real world. Our framework is robust to visual occlusion, electromagnetic interference, and contact common in many manipulation tasks by utilizing both inertial and visual information without relying on soft sensors that cannot distinguish deformation by motion from deformation by contact. Using our system, the human expert can perform a truly dexterous manipulation (e.g., pen spinning) which is impossible with bulky exoskeleton-type data acquisition approaches. We present samples of demonstration data for dexterous manipulation that are challenging with existing systems (e.g., visually occluded and contact-rich), which demonstrates the effectiveness and versatility of our framework.

Although only the hand motion data is shown in this work, we plan to integrate tactile measurements in the data by attaching tactile sensors (e.g., FSR) to the glove. Otherwise, a physics simulator [2, 3] can be used to estimate accurate contact forces to further enrich the data.

Acknowledgments

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